<table>
<tr><td colspan="2">Unit/Lesson Title Wave / Energy and mechanical waves</td><td>Lesson duration 120 min</td><td colspan="2">Stage 6 Year 11</td><td>Class/Group [redacted]</td></tr>
<tr><td colspan="2">Rationale

Students have prior exposure to notion of mechanical waves and how energy is transferred from Stage 5. From previous lessons in the unit, students have modelled oscillation of particles in transverse / longitudinal waves and used wave equation to solve problems</td><td colspan="2">Syllabus Outcomes

NESA syllabus outcomes, Year 11 Physics (2025)

Working Scientifically
PY-11WS-02
PY-11WS-03
PY11-WS-07
PY-11-02</td><td colspan="2">Syllabus Content</td></tr>
<tr><td>Learning Indicators/specific content to be taught

Students

• Identify that mechanical waves require a medium for propagation
• Explain how sound waves transfer energy through a medium

Students

• Conduct scientific investigation to collect data and information</td><td colspan="3">Learning Experience (what the teacher and learners will be doing)

Engage

Teacher reviews mechanical wave and shows again this animation of how particles oscillate in sound wave. Teacher introduces hands-on activity challenge:

Prove that sound wave transfers energy through medium

Explore

Students separate into groups of 3-4, each group given material:

• Tuning forks
• Ping-pong balls
• String
• Small speaker
• Paper
• Some amount of rice or sand</td><td colspan="2">Resources

Animation of particle motion in longitudinal wave.

Formative assessment (informal): groups’</td></tr>
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• Analyses and evaluates primary and secondary data and information	Groups are given 5-10 minutes to discuss and propose experiment design. Teacher records and use questioning to encourage the class to comment on them (e.g., striking tuning fork then touching it to ping-pong ball is not a valid design) and supplement where necessary (e.g., place ping-pong ball next to one tuning fork and strike another one with same frequency at a distance). Teacher should also acknowledge valid design using material not available in the room. Groups conduct the experiment using available material, then complete a lab report using template provided. **Explain** Teacher summarises groups' experiments then plays video clip: yidaki (digeridoo) playing levitating a piece of paper as another demonstration of energy transfer by sound wave. **Elaborate** Teacher plays two sci-fi movie space battle scenes then asks: *As a physicist, what do you think is wrong with these two scenes?* [Class discussion] Teacher allows students to freely exchange ideas about what are unphysical for about 3-5 minutes, acknowledge valid ideas (e.g., "hyperspace jump" is sci-fi not science) then focus on laser/explosion sound effects, "sound cannot travel through the vacuum of space" **Evaluate** [Exit ticket] Lesson ends with exit ticket about what can realistically be received by a spaceship during a space battle.	discussion and design ideas Formative assessment: lab report YouTube video of sound levitation Clip 1: Star Wars space battle scene (00:00 – 00:40) Clip 2: Star Trek space battle (00:40 – 01:16) Formative assessment: Exit ticket